

Notes: Xiong & Yan (RFS, 2010)

Heterogeneous Expectations and Bond Markets

Contents

1	Big picture	2
2	Model objects and measurement	2
2.1	Endowment economy and nominal short rate	2
2.2	Heterogeneous expectations	2
2.3	Econometrician's benchmark belief	3
2.4	Symmetric disagreement restriction	3
3	Equilibrium and bond pricing	3
3.1	Short rate, wealth ratio, and asset prices	3
3.2	Representative-investor belief	4
3.3	Bond prices in homogeneous economies	4
3.4	Bond prices in the heterogeneous economy	4
4	Mechanisms	4
4.1	Volatility amplification	4
4.2	Time-varying risk premia	5
4.3	Failure of the expectations hypothesis	5
4.4	Tent-shaped return-predictive factor	5
5	Calibration and simulation design	5
5.1	Baseline calibration	5
5.2	Empirical benchmarks matched by the simulation	5
6	Tables and figures	6
6.1	Figure 1 and Table 1: yield volatility and expectations-hypothesis failure	6
6.2	Figure 2 and Table 2: Cochrane–Piazzesi return predictability	7
7	Identification and interpretation	7
7.1	What is identified by the model	7
7.2	Why the representative-agent shortcut fails	7
7.3	Testable implication on trading volume	8
7.4	Main limitations	8
8	Short summary	8
9	Conclusion	8
9.1	Core conclusion	8
9.2	Economic intuition	8
9.3	Takeaway	9

1 Big picture

- **Question.** Can disagreement among investors explain why bond yields are so volatile, why the expectations hypothesis fails, and why a tent-shaped forward-rate factor predicts bond returns?
- **Core mechanism.** Two investor groups disagree about the long-run inflation target. They trade against each other; news shifts wealth toward the group whose belief is favored; the new wealth distribution changes bond prices.
- **Key state variable.** Bond prices depend not only on current inflation and average beliefs, but also on the *wealth-weighted* average belief of the two groups.
- **Volatility result.** Relative-wealth fluctuations amplify the response of yields to news. A modest amount of belief dispersion can make long yields much more volatile than in a representative-agent benchmark.
- **Risk-premium result.** From an econometrician's point of view, bond premia vary with the gap between the econometrician's belief and investors' wealth-weighted average belief.
- **Expectations-hypothesis result.** When the wealth-weighted average belief about future short rates is high, long bonds look cheap to the econometrician: yield spreads are high and future long yields tend to fall.
- **Cochrane–Piazzesi result.** The model produces a tent-shaped factor in forward rates because intermediate forward rates are most sensitive to beliefs about the long-run inflation target.
- **Economic takeaway.** Heterogeneous expectations are not just a small deviation from a representative-agent model. Trading and wealth redistribution make disagreement an endogenous state variable for bond prices and premia.

2 Model objects and measurement

2.1 Endowment economy and nominal short rate

The paper uses a Lucas-style endowment economy with one consumption good. Aggregate endowment satisfies

$$\frac{dD_t}{D_t} = \mu_D dt + \sigma_D dZ_D(t). \quad (1)$$

The price level evolves as

$$\frac{dp_t}{p_t} = \pi_t dt, \quad (2)$$

where π_t is inflation. Inflation mean-reverts to an unobservable inflation target θ_t :

$$d\pi_t = -\lambda_\pi(\pi_t - \theta_t)dt + \sigma_\pi dZ_\pi(t), \quad (3)$$

with

$$d\theta_t = -\lambda_\theta(\theta_t - \bar{\theta})dt + \sigma_\theta dZ_\theta(t). \quad (4)$$

Interpretation. The real side is deliberately simple. The asset-pricing action comes from how investors learn about θ_t , not from complicated consumption dynamics.

2.2 Heterogeneous expectations

Investors observe inflation and a signal flow S_t . In reality the signal is pure noise:

$$dS_t = dZ_S(t). \quad (5)$$

Group $i \in \{A, B\}$, however, believes the signal is correlated with the fundamental shock to θ_t :

$$dS_t = \phi_i dZ_\theta(t) + \sqrt{1 - \phi_i^2} dZ_S(t). \quad (6)$$

The group also may misperceive the volatility of θ_t by a factor k_i :

$$d\theta_t = -\lambda_\theta(\theta_t - \bar{\theta})dt + k_i\sigma_\theta dZ_\theta(t). \quad (7)$$

Interpretation. Disagreement is generated by different interpretations of the same public signal. The two groups see the same data but extract different news about long-run inflation.

2.3 Econometrician's benchmark belief

The econometrician knows that the signal is uninformative. Let $\hat{\theta}_t^R$ denote the econometrician's posterior mean. In the steady state,

$$d\hat{\theta}_t^R = -\lambda_\theta(\hat{\theta}_t^R - \bar{\theta})dt + \frac{\lambda_\pi}{\sigma_\pi}\bar{v}d\hat{Z}_\pi^R(t), \quad (8)$$

where $\bar{v}$ is the stationary posterior variance and $d\hat{Z}_\pi^R$ is the information shock in inflation.

Interpretation. The econometrician's belief is the paper's benchmark for judging whether equilibrium bond prices appear cheap or expensive.

2.4 Symmetric disagreement restriction

The baseline calibration imposes

$$\phi_A = -\phi_B = \phi, \quad k_A = k_B = \frac{1}{\sqrt{1 - \phi^2}}. \quad (9)$$

Under these restrictions, the two groups' belief difference follows

$$d(\hat{\theta}_t^A - \hat{\theta}_t^B) = -\left(\lambda_\theta + \frac{\lambda_\pi^2}{\sigma_\pi^2}\bar{v}\right)(\hat{\theta}_t^A - \hat{\theta}_t^B)dt + \frac{2\phi\sigma_\theta}{\sqrt{1 - \phi^2}}dS_t, \quad (10)$$

and if the initial average belief equals the econometrician's belief, then

$$\frac{1}{2}(\hat{\theta}_t^A + \hat{\theta}_t^B) = \hat{\theta}_t^R. \quad (11)$$

Interpretation. This restriction isolates the pure disagreement channel. The average belief is not biased relative to the econometrician, and the two groups have the same posterior variance. The remaining effect comes from trading and wealth redistribution.

3 Equilibrium and bond pricing

3.1 Short rate, wealth ratio, and asset prices

Let $\eta_t = W_t^B/W_t^A$ be the wealth ratio. The equilibrium nominal short rate is

$$r_t = \pi_t + \rho + \mu_D - \sigma_D^2. \quad (12)$$

The wealth ratio evolves as

$$d\ln(\eta_t) = \frac{\lambda_\pi}{\sigma_\pi}(\hat{\theta}_t^B - \hat{\theta}_t^A)\left[\hat{\theta}_t^R - \frac{\hat{\theta}_t^A + \hat{\theta}_t^B}{2}\right]dt + \frac{\lambda_\pi}{\sigma_\pi}(\hat{\theta}_t^B - \hat{\theta}_t^A)d\hat{Z}_\pi^R(t). \quad (13)$$

For a nominal asset that pays X_T once at time T , the equilibrium price is

$$P_X(t) = \omega_t^A P_X^A(t) + \omega_t^B P_X^B(t), \quad (14)$$

where ω_t^i is group i 's wealth share and $P_X^i(t)$ is the price in a homogeneous economy populated only by group i .

Interpretation. The price is a wealth-weighted average of the two groups' valuations. This is the central aggregation result. Disagreement matters because the weights are endogenous and move with shocks.

3.2 Representative-investor belief

A representative investor can reproduce equilibrium prices only if that investor's belief is

$$\hat{\theta}_t^N = \omega_t^A \hat{\theta}_t^A + \omega_t^B \hat{\theta}_t^B. \quad (15)$$

Interpretation. The representative agent's belief is not a normal Bayesian posterior with a fixed prior. It moves with wealth shares. This is why a representative-agent model misses the underlying economic mechanism.

3.3 Bond prices in homogeneous economies

In a homogeneous economy populated only by group i , the price of a zero-coupon nominal bond with maturity τ is

$$B^H(\tau, \pi_t, \hat{\theta}_t^i) = \exp\{-a_\pi(\tau)\pi_t - a_\theta(\tau)\hat{\theta}_t^i - b(\tau)\}, \quad (16)$$

where

$$a_\pi(\tau) = \frac{1}{\lambda_\pi}(1 - e^{-\lambda_\pi \tau}), \quad (17)$$

$$a_\theta(\tau) = \frac{1}{\lambda_\theta}(1 - e^{-\lambda_\theta \tau}) + \frac{1}{\lambda_\pi - \lambda_\theta}(e^{-\lambda_\pi \tau} - e^{-\lambda_\theta \tau}). \quad (18)$$

Interpretation. The loading on current inflation declines with maturity. The loading on the inflation target belief is hump-shaped when θ_t mean-reverts, so intermediate-maturity yields and forward rates are especially sensitive to beliefs.

3.4 Bond prices in the heterogeneous economy

The heterogeneous-economy bond price is

$$B_t = \omega_t^A B^H(\tau, \pi_t, \hat{\theta}_t^A) + \omega_t^B B^H(\tau, \pi_t, \hat{\theta}_t^B). \quad (19)$$

The implied yield is

$$Y_t(\tau) = \frac{a_\pi(\tau)}{\tau}\pi_t + \frac{b(\tau)}{\tau} - \frac{1}{\tau} \log \left(\omega_t^A e^{-a_\theta(\tau)\hat{\theta}_t^A} + \omega_t^B e^{-a_\theta(\tau)\hat{\theta}_t^B} \right). \quad (20)$$

Interpretation. Bond yields are non-affine because they aggregate two exponential valuations. The nonlinearity comes from belief heterogeneity and market clearing.

4 Mechanisms

4.1 Volatility amplification

- If group A is more optimistic about future rates, it takes positions that profit when rates rise.
- A positive information shock transfers wealth toward group A.
- Group A's belief receives a larger weight in equilibrium prices.
- The initial shock therefore has a larger price impact than it would in a homogeneous economy.

Interpretation. The model creates a feedback channel: news changes prices directly and also changes the wealth distribution, which changes whose beliefs matter more.

4.2 Time-varying risk premia

From the econometrician's point of view, the real stochastic discount factor satisfies

$$\frac{dM_t}{M_t} = -(\rho + \mu_D - \sigma_D^2)dt - \sigma_D dZ_D - \frac{\lambda_\pi}{\sigma_\pi} \left(\hat{\theta}_t^R - \sum_{i=A,B} \omega_t^i \hat{\theta}_t^i \right) d\hat{Z}_\pi^R. \quad (21)$$

Interpretation. The market price of the inflation-information shock is proportional to the difference between the econometrician's belief and investors' wealth-weighted average belief. Since wealth shares move endogenously, risk premia vary over time.

4.3 Failure of the expectations hypothesis

The expectations hypothesis predicts that a positive yield spread should be followed by rising long yields. In the model, a high wealth-weighted belief about future short rates has two effects:

- Investors discount long bonds more heavily, so long yields and yield spreads are high.
- From the econometrician's perspective, long bonds look cheap, so future long-bond prices are expected to rise and yields are expected to fall.

Interpretation. The same belief state creates both a high yield spread and expected future yield declines. This reproduces the classic Campbell–Shiller rejection pattern.

4.4 Tent-shaped return-predictive factor

The forward rate for loans beginning τ years ahead can be written approximately as

$$F_t(\tau) = a'_\pi(\tau)\pi_t + b'(\tau) + a'_\theta(\tau)\tilde{\theta}_t, \quad (22)$$

where $\tilde{\theta}_t$ is a bond-price-weighted average of the two groups' beliefs. Because $a'_\theta(\tau)$ is hump-shaped, intermediate forward rates respond more strongly to belief changes than short or distant forward rates.

Interpretation. A high wealth-weighted belief raises the three-year forward rate more than the one-year and five-year rates. This makes a tent-shaped combination of forward rates a natural predictor of bond excess returns.

5 Calibration and simulation design

5.1 Baseline calibration

The baseline parameter values are

$$\lambda_\theta = 0.07, \quad \lambda_\pi = 1, \quad \sigma_\pi = 1.25\%, \quad \sigma_\theta = 1.2\%, \quad (23)$$

$$\mu_D = 2\%, \quad \sigma_D = 2\%, \quad \phi = 0.75, \quad \rho = 2\%, \quad (24)$$

$$\omega_0^A = \omega_0^B = 0.5, \quad \pi_0 = \theta_0 = \hat{\theta}_0^A = \hat{\theta}_0^B = \bar{\theta} = 2\%. \quad (25)$$

The simulation runs the economy for 50 years at daily frequency, extracts yields and forward rates at monthly frequency, and repeats the experiment 10,000 times.

Interpretation. The disagreement is modest: the paper reports an average dispersion between group beliefs of only about 1.70%. This is important because the main results do not require extreme belief differences.

5.2 Empirical benchmarks matched by the simulation

- **Bond-yield volatility.** Long yields are too volatile in standard representative-agent models. The heterogeneous model amplifies long-yield volatility.

- **Campbell–Shiller regressions.** The model generates negative yield-change regression coefficients at long maturities, consistent with the failure of the expectations hypothesis.
- **Cochrane–Piazzesi regressions.** The model generates a single tent-shaped forward-rate factor that predicts two- to five-year excess bond returns.

Interpretation. The calibration is not an estimation exercise. It is a disciplined model experiment showing that the disagreement mechanism can quantitatively mimic several bond-market facts.

6 Tables and figures

6.1 Figure 1 and Table 1: yield volatility and expectations-hypothesis failure

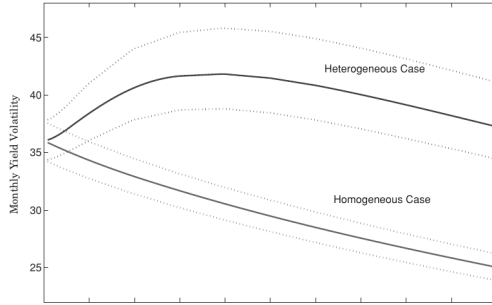


Figure 1: term structure of monthly bond-yield volatility.

Table 1
Coefficients of yield change regressions

n	2	3	6	12	24	48	120
Panel A: Results from Campbell–Lo–MacKinlay							
β_n	0.003	-0.145	-0.835	-1.435	-1.448	-2.262	-4.226
<i>s.e.</i>	(0.191)	(0.282)	(0.442)	(0.599)	(1.004)	(1.458)	(2.076)
Panel B: Results from our simulation of a heterogeneous economy							
β_n	-1.037	-1.104	-1.304	-1.699	-2.435	-3.231	-3.499
<i>s.e.</i>	(0.516)	(0.541)	(0.620)	(0.785)	(1.110)	(1.632)	(1.986)

Table 1: coefficients in Campbell–Shiller-style yield-change regressions.

Read:

- In Figure 1, the heterogeneous economy produces a hump-shaped volatility curve: volatility starts near 36 basis points per month, rises above 41 basis points around three to four years, and remains high at long maturities.
- The homogeneous benchmark declines monotonically from about 36 to about 25 basis points per month by ten years.
- The paper reports that the 10-year volatility amplification is about 12 basis points per month, roughly one-third of total 10-year yield volatility.
- Table 1 compares the Campbell–Shiller empirical pattern with the simulation. In the data, β_n falls from 0.003 for two months to -4.226 for 120 months. In the simulation, it falls from -1.037 to -3.499.
- The expectations-hypothesis null is $\beta_n = 1$. Both the empirical and simulated long-maturity coefficients move in the opposite direction.

Economic message: Heterogeneous expectations create both high long-yield volatility and negative yield-change predictability. The mechanism is not an exogenous risk-premium process; it comes from wealth moving toward whichever belief is favored by the news.

6.2 Figure 2 and Table 2: Cochrane–Piazzesi return predictability

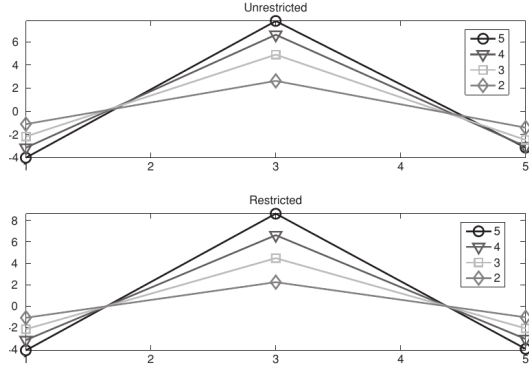


Figure 2: unrestricted and restricted Cochrane–Piazzesi coefficients.

Table 2
Coefficients in Cochrane–Piazzesi bond return regression

	Panel A: Unrestricted				Panel B: Restricted			
	$\beta_1(n)$	$\beta_3(n)$	$\beta_5(n)$	R^2	$\beta_1(n)$	$\beta_3(n)$	$\beta_5(n)$	R^2
$rx(2)$	-1.11	2.63	-1.42	20.8%	-1.07	2.24	-1.03	20.7%
<i>s.e.</i>	(0.05)	(0.38)	(0.38)		(0.05)	(0.40)	(0.40)	
$rx(3)$	-2.18	4.91	-2.48	20.1%	-2.13	4.48	-2.06	20.0%
<i>s.e.</i>	(0.09)	(0.79)	(0.78)		(0.09)	(0.80)	(0.79)	
$rx(4)$	-3.15	6.63	-3.05	19.5%	-3.15	6.62	-3.05	19.5%
<i>s.e.</i>	(0.14)	(1.18)	(1.18)		(0.14)	(1.18)	(1.17)	
$rx(5)$	-4.02	7.82	-3.15	19.1%	-4.11	8.64	-3.97	19.1%
<i>s.e.</i>	(0.18)	(1.56)	(1.56)		(0.18)	(1.54)	(1.53)	

Table 2: simulated Cochrane–Piazzesi regression coefficients.

Read:

- Figure 2 shows the tent shape: the coefficient on the three-year forward rate is positive and large, while coefficients on the one-year yield and five-year forward rate are negative.
- Longer-maturity excess returns have larger loadings on the same tent-shaped factor.
- Table 2 reports R^2 values around 19–21% across two- to five-year excess-return regressions.
- The restricted single-factor regressions have almost the same R^2 values as the unrestricted regressions, suggesting that one common factor captures most of the return-predictive content.
- For example, in the restricted five-year return regression, the coefficients are approximately -4.11, 8.64, and -3.97, with $R^2 = 19.1\%$.

Economic message: The CP factor is interpretable as a proxy for the market’s wealth-weighted belief about future short rates. Because intermediate forward rates load most on the inflation-target belief, the predictive factor naturally has a tent shape.

7 Identification and interpretation

7.1 What is identified by the model

- The model isolates a channel: disagreement $\rightarrow$ speculative trading $\rightarrow$ relative-wealth changes $\rightarrow$ bond-price and risk-premium dynamics.
- The symmetric learning setup rules out two alternative explanations: biased average beliefs and underestimation of posterior variance.
- The calibration asks whether a modest disagreement level can generate magnitudes similar to observed bond-market puzzles.

Interpretation. This is a theoretical identification strategy, not an empirical causal design. The paper shows that the mechanism is internally coherent and quantitatively plausible.

7.2 Why the representative-agent shortcut fails

- A representative agent can always be constructed in complete markets, but its belief must equal the wealth-weighted average belief.
- This belief changes when wealth shares change, even if the underlying inflation data are unchanged.
- Therefore, the representative-agent belief does not follow a standard Bayesian updating process.

Interpretation. The representative-agent version hides the actual source of price movements. The state variable that matters is created by heterogeneous trading.

7.3 Testable implication on trading volume

The paper also shows that trading volume rises with belief dispersion. Higher belief dispersion implies larger speculative positions, greater wealth-share volatility, and more rebalancing. Combined with the yield results, the model predicts that higher belief dispersion should be associated with higher trading volume, higher bond-yield volatility, and a flatter yield curve.

Interpretation. This connects the pricing model to market activity. Disagreement should show up not only in prices but also in volume.

7.4 Main limitations

- The model uses logarithmic utility and complete markets to obtain closed-form aggregation.
- Disagreement is imposed through perceived signal informativeness rather than estimated from survey data.
- The calibration is designed to match major patterns, but the paper does not estimate the full model by maximum likelihood or GMM.
- The model focuses on nominal bond markets and disagreement about inflation targets; real-side disagreement is discussed as conceptually similar but not central in the paper.

Interpretation. The assumptions make the mechanism transparent. The tradeoff is that the paper is more a proof of a powerful economic channel than a fully estimated term-structure model.

8 Short summary

- Investors disagree about the long-run inflation target because they interpret a public signal differently.
- The average of the two groups' beliefs can be kept equal to the econometrician's belief, so the analysis isolates disagreement rather than average optimism or pessimism.
- Disagreement induces speculative positions. News transfers wealth between groups, and wealth shifts change the weights placed on each group's bond valuation.
- This relative-wealth channel amplifies bond-yield volatility, especially at longer maturities.
- The same channel produces time-varying risk premia and can explain why high yield spreads are followed by falling long yields.
- Intermediate forward rates are most sensitive to beliefs about the inflation target, so the model generates a tent-shaped return-predictive factor.
- The simulation uses modest belief dispersion and matches the qualitative and quantitative direction of several famous bond-market puzzles.

9 Conclusion

9.1 Core conclusion

Heterogeneous expectations can explain several major bond-market anomalies because disagreement changes the distribution of wealth, and the distribution of wealth determines whose beliefs are reflected in prices.

9.2 Economic intuition

When a group is optimistic about future short rates, it bets accordingly. If news supports that view, the group gains wealth. Its belief then receives more weight in equilibrium pricing. This feedback makes yields more volatile and creates time-varying premia.

9.3 Takeaway

The paper's contribution is to convert belief heterogeneity from a static dispersion variable into a dynamic pricing state variable. In bond markets, the important object is not only what investors believe, but also how much wealth is controlled by each belief group.